Android based Crowdsourced Disaster Alert & Safety App

Design Document

Version 1.0

Group Id: F25PROJECT384C7

Supervisor Name : Irshad Nasir

Revision History

Date (dd/mm/yyyy)	Version	Description	Author
12-01-2026	1.0	Natural disasters such as floods, earthquakes, and fires can strike unexpectedly, putting lives and property at risk. Timely information during such events is crucial, yet traditional communication channels are often slow or unreliable. The **Crowdsourced Disaster Alert & Safety App (SoSync)** is an Android-based mobile application designed to address this challenge. It enables users to report disasters in real time, receive instant alerts based on their location, and verify reports collectively. By integrating features like interactive maps, SOS alerts, and an admin verification system, the app leverages community participation and cloud technologies to enhance disaster preparedness, response, and overall public safety.	bc240222825

Table of Contents

1. Introduction of Design Document

2. Entity Relationship Diagram (ERD)

3. Sequence Diagrams

4. Architecture Design Diagram

5. Class Diagram

6. Database Design

7. Interface Design

8. Test Cases

1. Introduction of Design Document

The **Design Document** for the **Crowdsourced Disaster Alert & Safety App (Sosync)** serves as a detailed blueprint that transforms the Software Requirements Specification (SRS) into a clear and structured plan for implementation. This document explains how the system will be built, how its components will interact, and how data will be managed throughout the application. The primary purpose of this design phase is to provide developers, testers, and stakeholders with a shared understanding of the system's architecture, functionality, and behavior before the actual coding process begins. By defining technical and logical structures in advance, the design document minimizes ambiguity and ensures that development proceeds in an organized and efficient manner.

This document includes several essential design components that collectively define the complete system.

The **Architecture Design** presents the overall structure of the application, highlighting the use of an **Expo-based React Native mobile application** as the front end. Expo is selected to simplify development, testing, and deployment while ensuring cross-platform compatibility. The backend primarily utilizes **Firebase services** such as Authentication, Firestore or Realtime Database, Storage, and Cloud Messaging to support real-time reporting, media handling, and notifications. If additional business logic, advanced processing, or third-party integrations are required, a **Node.js backend** may be introduced to handle server-side operations and APIs.

The **Entity Relationship Diagram (ERD)** presents a clear representation of the system's data model by illustrating entities such as users, disaster reports, votes, SOS alerts, and admins, along with their relationships. This helps in organizing data efficiently, reducing redundancy, and maintaining data integrity. A well-structured ERD supports smooth data retrieval and ensures that the system can handle large volumes of real-time data during emergencies.

The **Use Case Diagrams and Sequence Diagrams** describe how users and admins interact with the system over time. These diagrams clearly define workflows such as user registration, disaster reporting, community voting, SOS

activation, and admin verification. By visualizing step-by-step interactions, developers gain insight into system behavior, making it easier to implement accurate and coordinated logic.

The **Class Diagram** provides an object-oriented view of the system, identifying major classes such as User, DisasterReport, VoteManager, NotificationService, and SOSManager. This modular structure improves code maintainability and allows future enhancements without disrupting existing functionality.

The **Database Design** section defines collections, fields, and relationships used in Firebase, ensuring efficient storage, real-time updates, and secure access control.

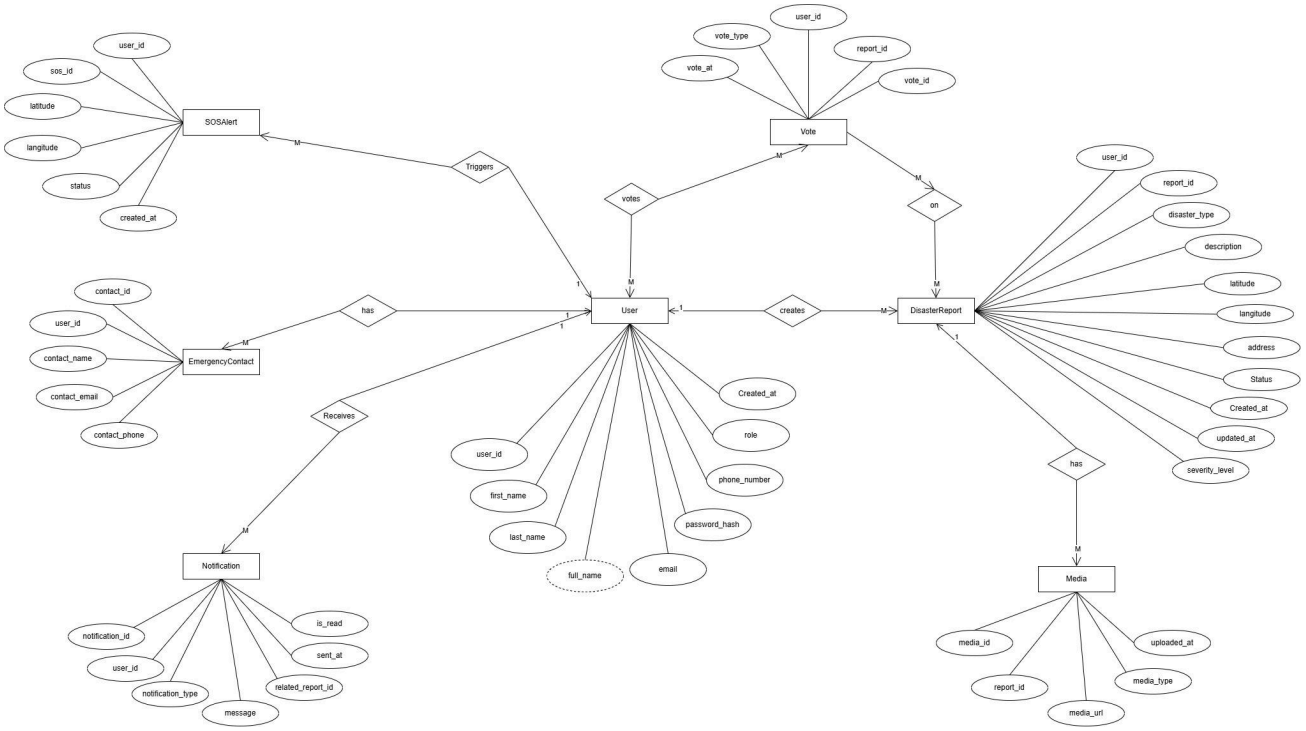
The **Interface Design** section focuses on user experience, detailing key screens such as login, report creation, map view, report details, **SOS** screen, and admin dashboard, with an emphasis on simplicity and quick access during emergencies.

The **Test Cases and Validation Strategy** ensure that all functional and non-functional requirements are met. Testing covers authentication, reporting, voting, notifications, SOS functionality, performance, and security.

The **design phase** plays a critical role in the software development lifecycle by identifying potential risks early, reducing development cost, and improving collaboration.

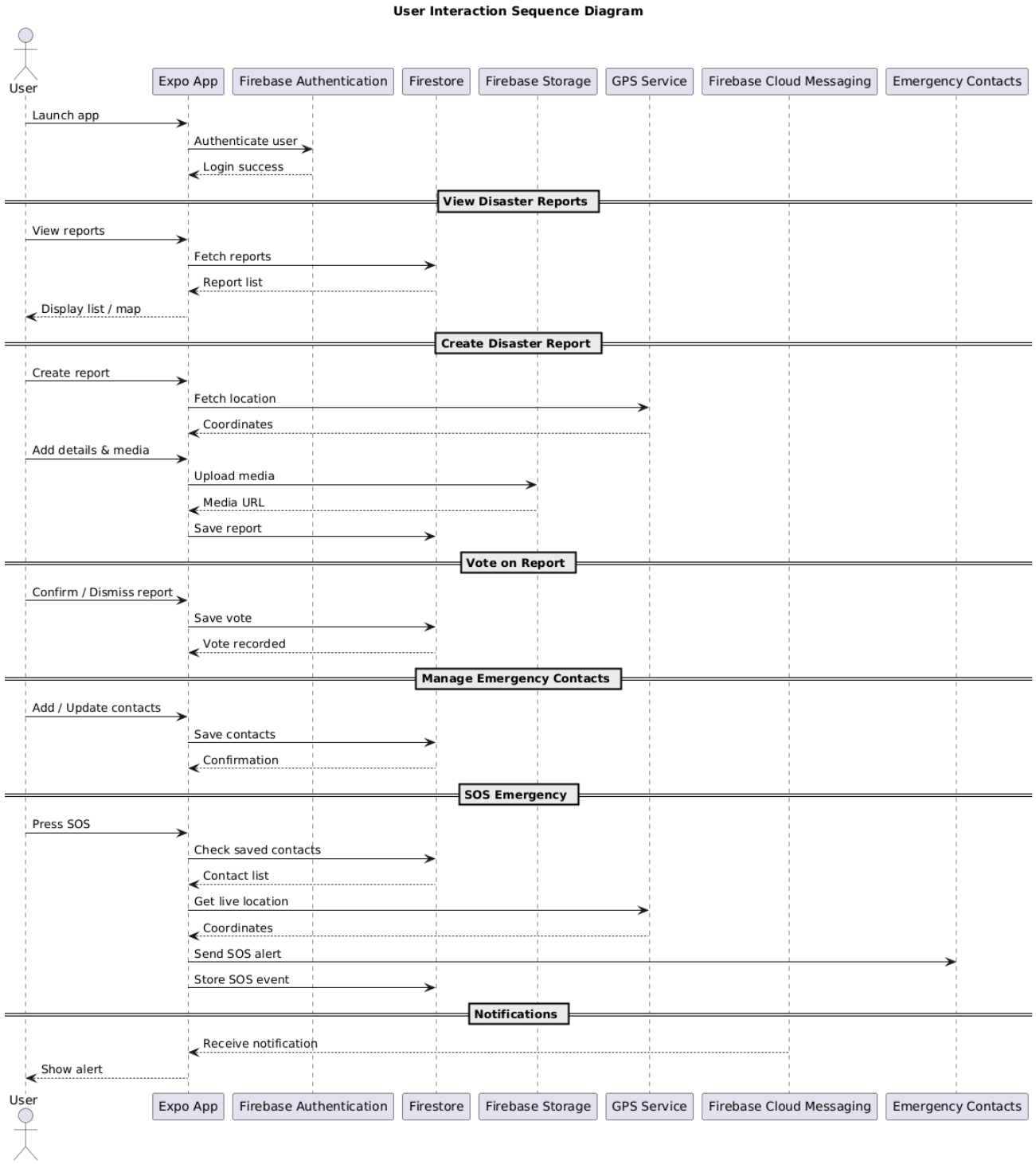
This **Design Document** serves as a solid foundation for building a reliable, scalable, and user-centered disaster alert system that enhances public safety and community resilience.

2. Entity Relationship Diagram (ERD) (To be developed using Microsoft Visio or any other drawing software of your choice)

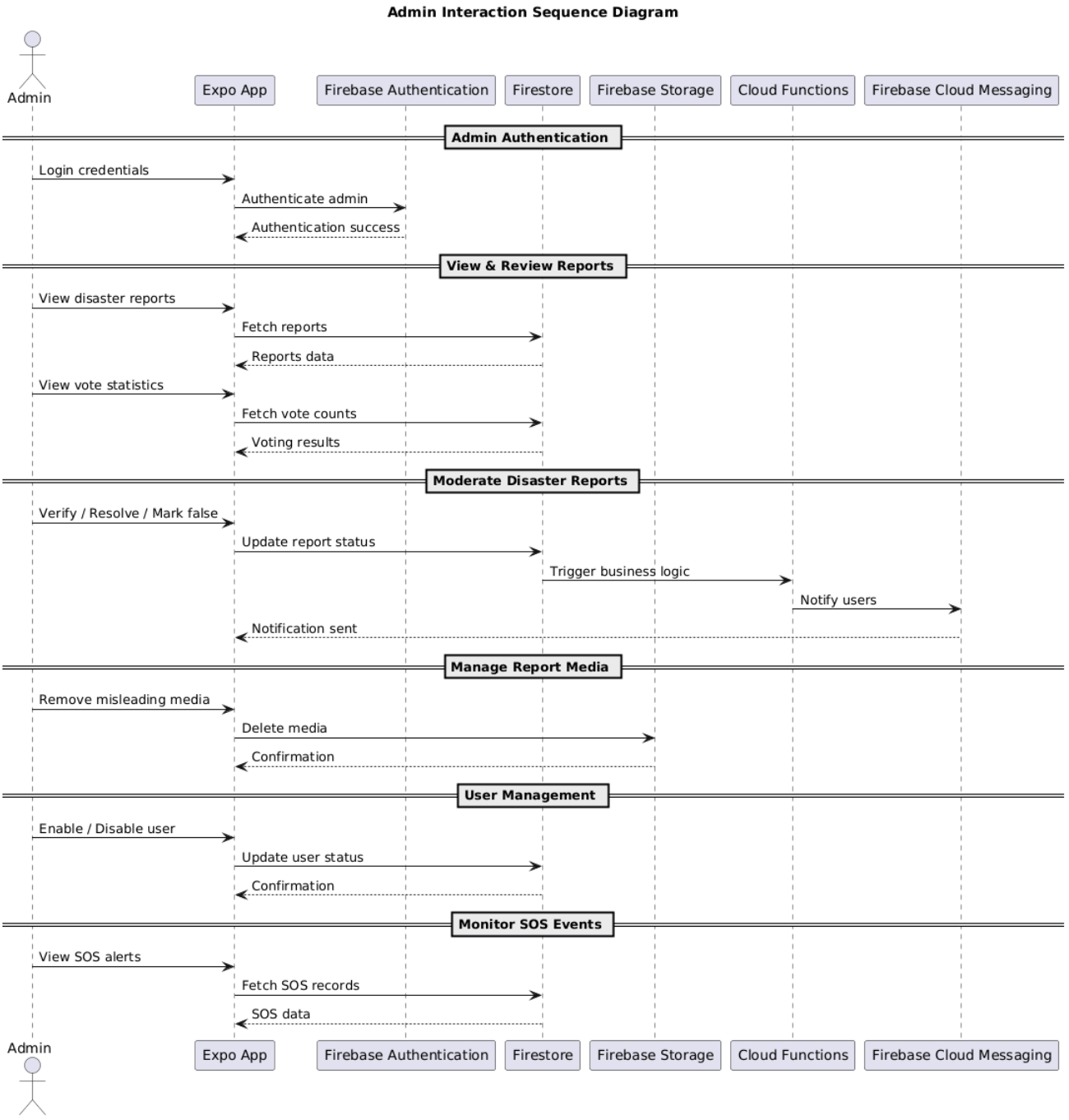


3. Sequence Diagrams (To be developed using Rational Rose or any other drawing software of your choice)

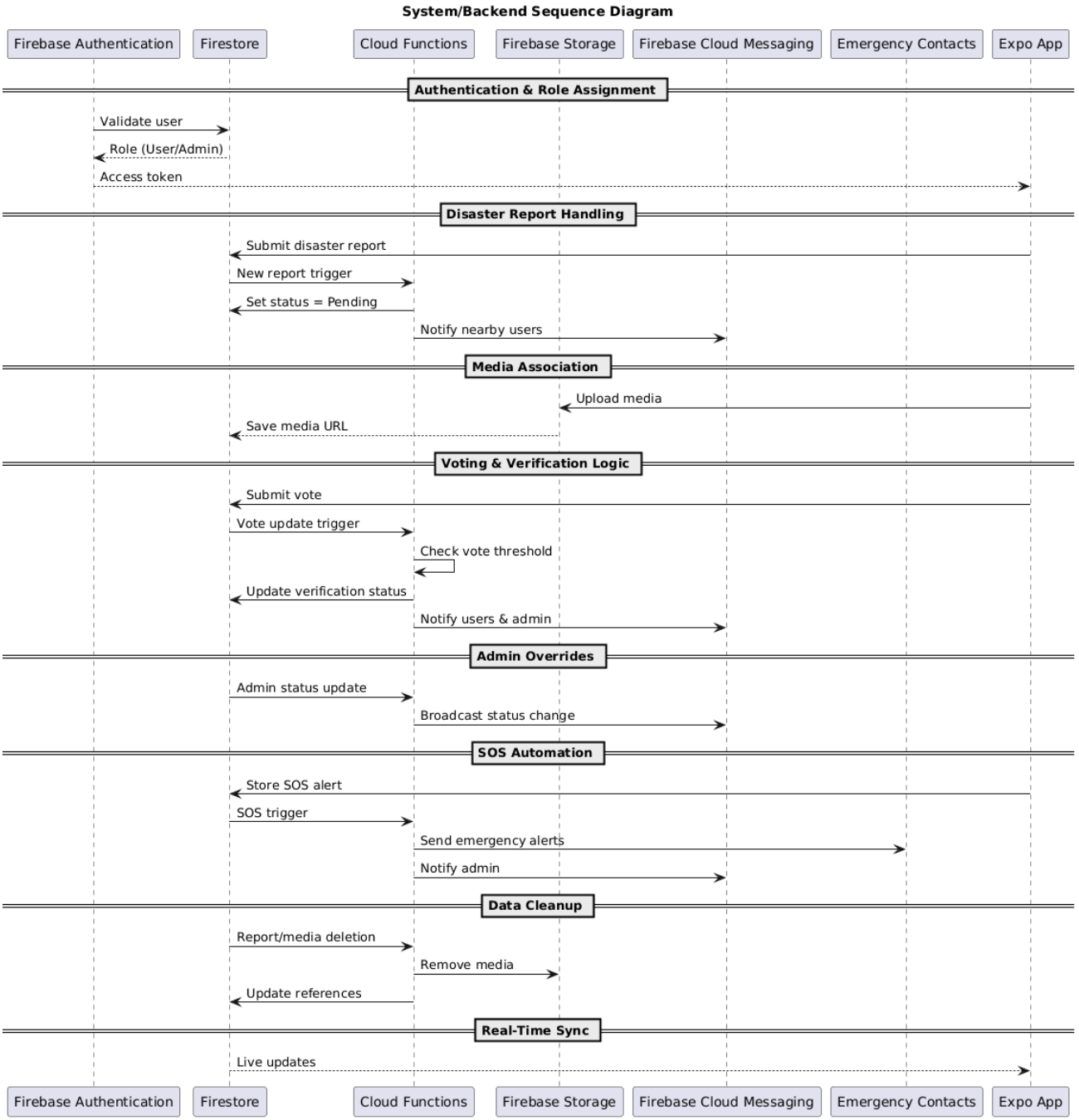
User Diagram



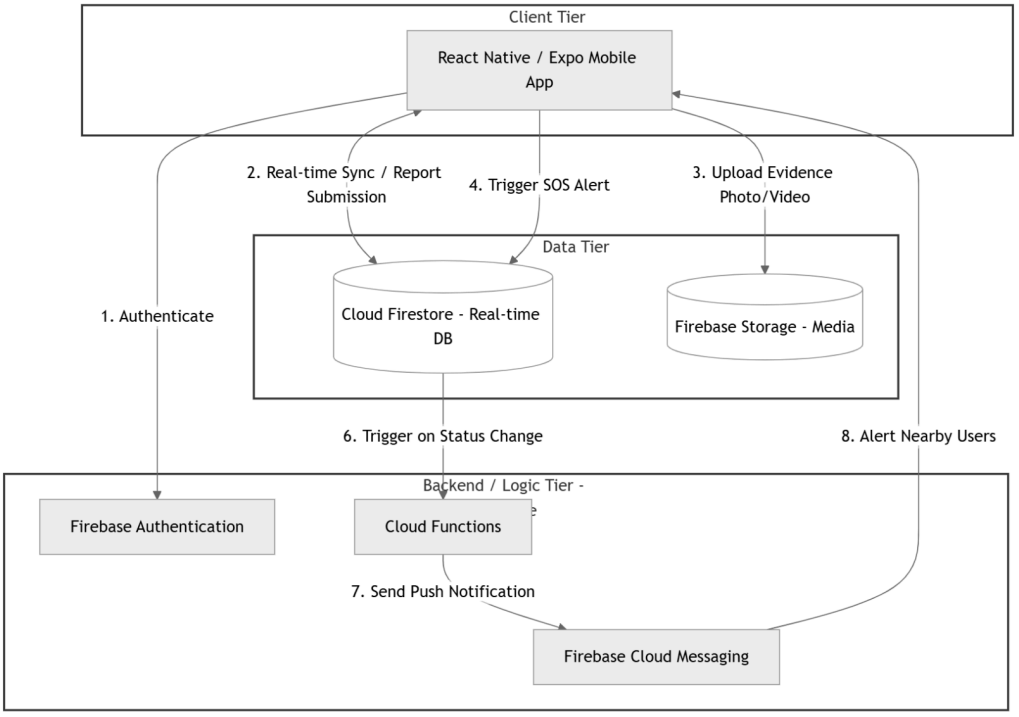
Admin diagram

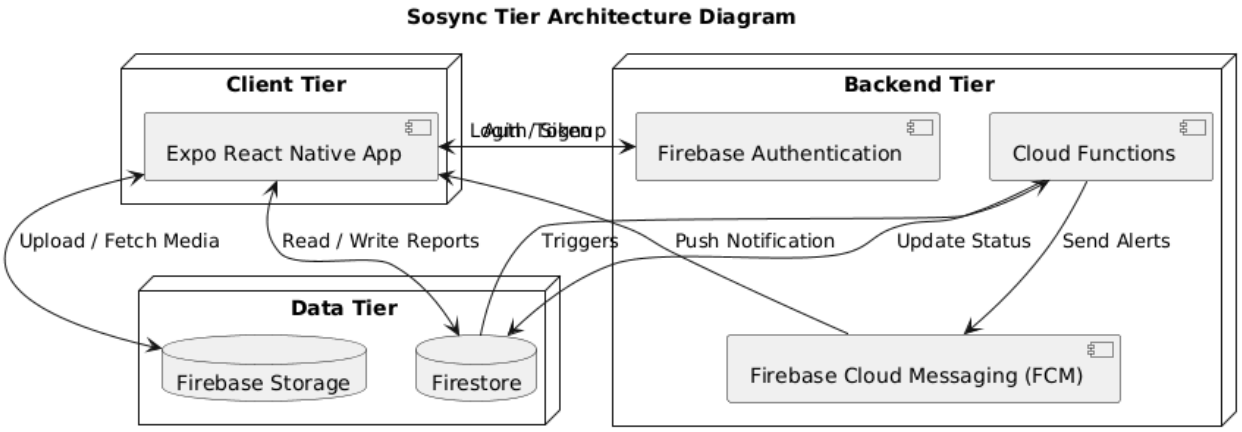


System Diagram

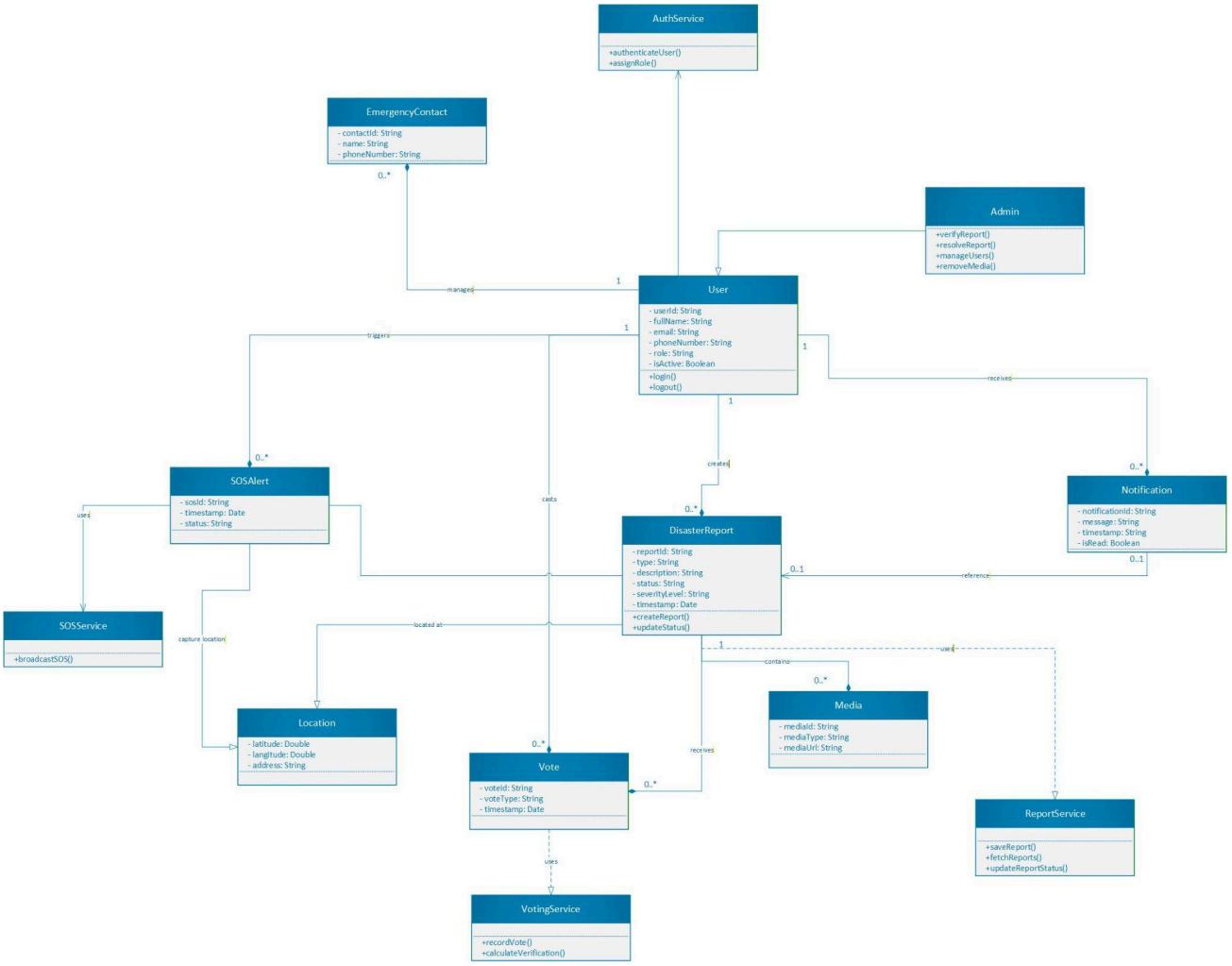


4. Architecture Design Diagram

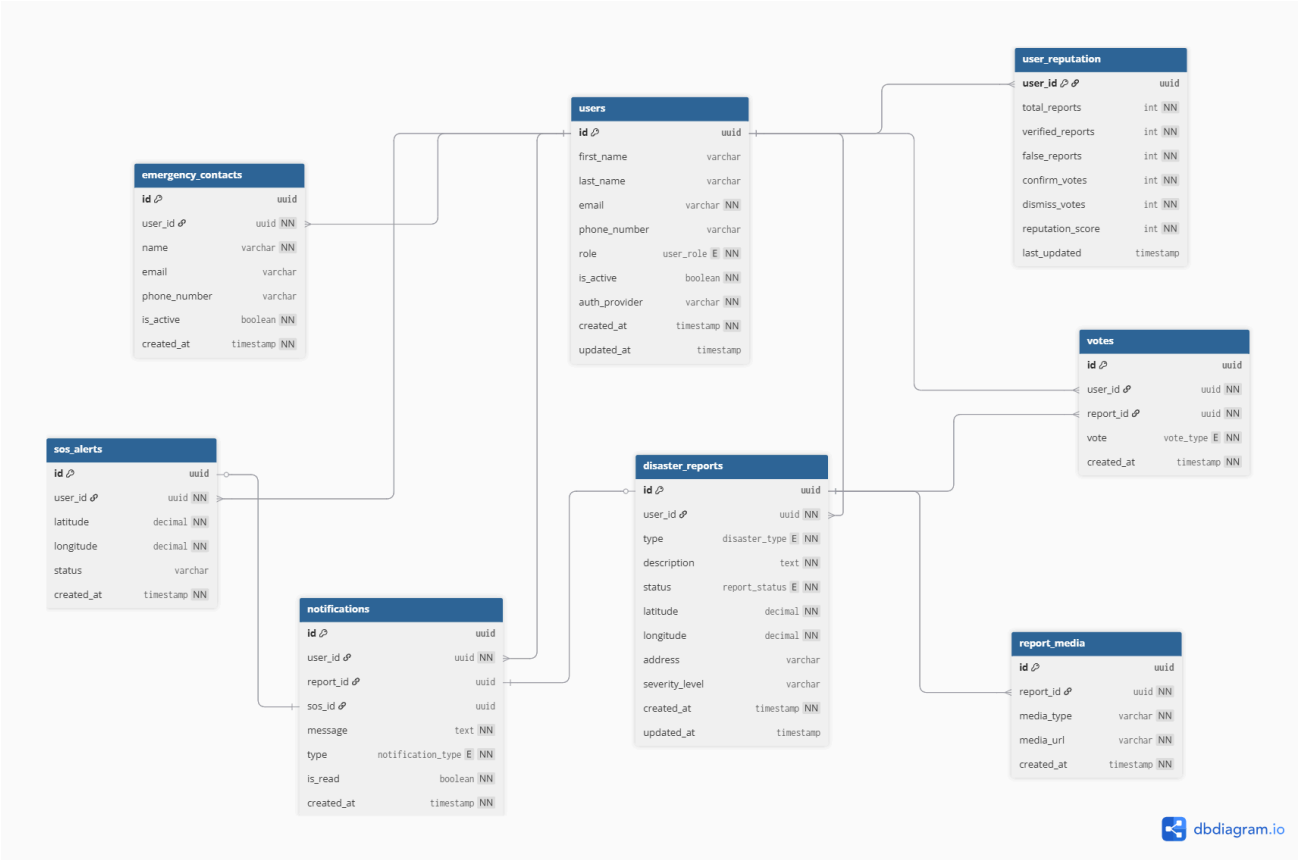




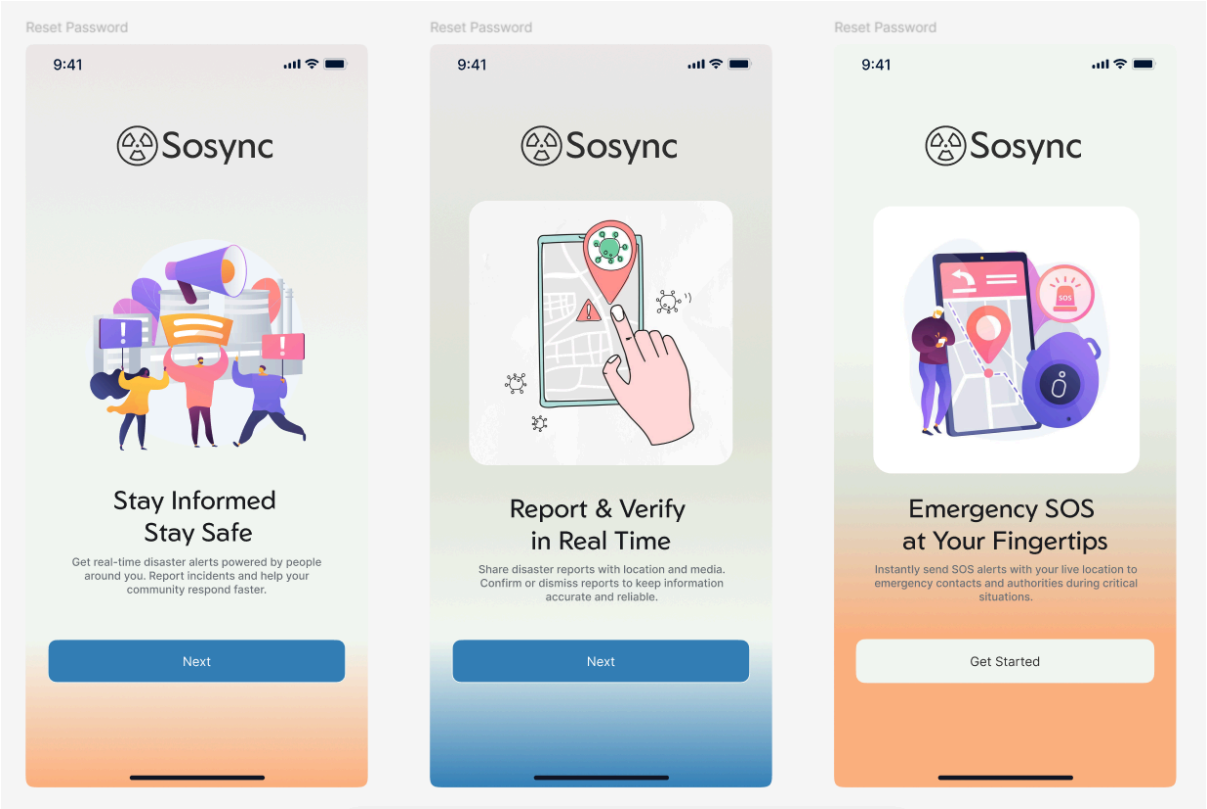
5. Class Diagram

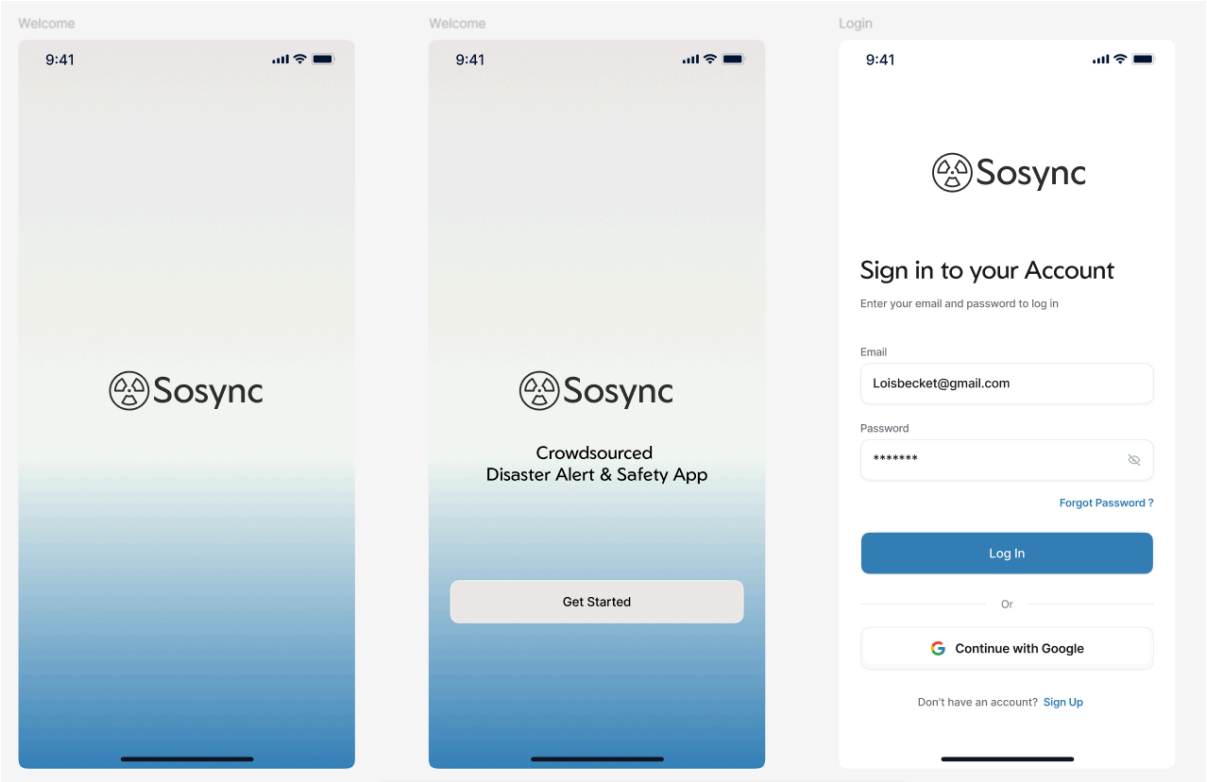


6. Database Design



7. Interface Design





Sign Up

9:41

Sosync

Sign up

Create an account to continue!

First Name

Lois

Last Name

Becket

Email

Loisbecket@gmail.com

Password

****

Register

Or

Continue with Google

Already have an account? Login

Forget Password

9:41

Sosync

Forget Password

Enter your email to access your account.

Email

Loisbecket@gmail.com

Proceed

Already have an account? Login

8. Test Cases

Test Case: verify that a user can register

Field	Details
Test Case ID	TC-01
Use Case Name	UC-01 – User Registration (Email)
Description	Verify that a new user can register using a valid email and password
Input	Valid email, valid password
Expected Output	User account is created and user is logged in
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: verify that user can login

Field	Details
Test Case ID	TC-02
Use Case Name	UC-02 – User Login (Email)
Description	Verify login using registered email and password
Input	Registered email, correct password
Expected Output	User authenticated and redirected to dashboard
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test Case: Login using Google Auth

Field	Details
Test Case ID	TC-03
Use Case Name	UC-03 – User Login (Google)

Description	Verify login using Google authentication
Input	Google account credentials
Expected Output	User logged in and profile created if first-time login
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: verify that user can logout

Field	Details
Test Case ID	TC-04
Use Case Name	UC-04 – User Logout
Description	Verify that user can log out successfully
Input	Logout request
Expected Output	User session terminated and redirected to login screen
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: verify that user can update profile

Field	Details
Test Case ID	TC-05
Use Case Name	UC-05 – Update User Profile
Description	Verify that user can update profile information
Input	Updated name / phone number
Expected Output	Profile data updated successfully
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: Add emergency contact

Field	Details
Test Case ID	TC-06
Use Case Name	UC-06 – Add Emergency Contact
Description	Verify user can add an emergency contact
Input	Contact name, phone or email
Expected Output	Emergency contact saved successfully
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: Update / Delete Emergency Contact

Field	Details
Test Case ID	TC-07
Use Case Name	UC-07 – Update/Delete Emergency Contact
Description	Verify user can update or delete emergency contacts
Input	Updated contact details or delete request
Expected Output	Contact updated or removed successfully
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test Case: Create Disaster Report

Field	Details

Test Case ID	TC-08
Use Case Name	UC-08 – Create Disaster Report
Description	Verify user can submit a disaster report
Input	Disaster type, description, GPS location
Expected Output	Report saved with status **PENDING**
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: Upload Media to Disaster Report

Field	Details
Test Case ID	TC-09
Use Case Name	UC-09 – Upload Media
Description	Verify user can upload image/video with report
Input	Image or video file
Expected Output	Media uploaded and linked to report
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test Case: View Disaster Reports (List / Map)

Field	Details
Test Case ID	TC-10
Use Case Name	UC-10 – View Disaster Reports
Description	Verify reports are viewable in list and map views
Input	View reports request
Expected Output	Reports displayed with location and status
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: Vote on Disaster Report

Field	Details
Test Case ID	TC-11
Use Case Name	UC-11 – Vote on Disaster Report
Description	Verify user can confirm or dismiss a report
Input	Vote type (Confirm / Dismiss)
Expected Output	Vote recorded and affects verification logic
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: Send SOS Alert

Field	Details
Test Case ID	TC-12
Use Case Name	UC-12 – Send SOS Alert
Description	Verify SOS alert can be sent with live location
Input	SOS button press, GPS location
Expected Output	SOS stored and alerts sent to contacts
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: View Notifications

Field	Details
Test Case ID	TC-13
Use Case Name	UC-13 – View Notifications

Description	Verify user can view received notifications
Input	Open notifications screen
Expected Output	Notifications displayed correctly
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: Mark Notification as Read

Field	Details
Test Case ID	TC-14
Use Case Name	UC-14 – Mark Notification as Read
Description	Verify user can mark notifications as read
Input	Notification selected
Expected Output	Notification marked as read
Result (Pass/Fail)	Pass
Tested By / Actor	User

Test case: Admin Login

Field	Details
Test Case ID	TC-15
Use Case Name	UC-15 – Admin Login
Description	Verify that an admin can log in with valid credentials
Input	Admin email, correct password
Expected Output	Admin authenticated and redirected to admin dashboard
Result (Pass/Fail)	Pass
Tested By / Actor	Admin

Test case: View All Disaster Reports (Admin)

Field	Details
Test Case ID	TC-16
Use Case Name	UC-16 – View All Disaster Reports
Description	Verify admin can view all disaster reports
Input	Request to view reports
Expected Output	All reports displayed with details and status
Result (Pass/Fail)	Pass
Tested By / Actor	Admin

Test case: Verify Disaster Report

Field	Details
Test Case ID	TC-17
Use Case Name	UC-17 – Verify Disaster Report
Description	Verify admin can mark a report as verified
Input	Report ID, status = VERIFIED
Expected Output	Report status updated and notifications sent
Result (Pass/Fail)	Pass
Tested By / Actor	Admin

Test case: Mark Disaster Report as Resolved

Field	Details
Test Case ID	TC-18
Use Case Name	UC-18 – Resolve Disaster Report
Description	Verify admin can mark a disaster report as resolved
Input	Report ID, status = RESOLVED
Expected Output	Report marked resolved and users notified
Result (Pass/Fail)	Pass
Tested By / Actor	Admin

Test case: Mark Disaster Report as False Alarm

Field	Details
Test Case ID	TC-19
Use Case Name	UC-19 – Mark False Alarm
Description	Verify admin can mark a report as false alarm
Input	Report ID, status = FALSE_ALARM
Expected Output	Report updated and reputation adjusted
Result (Pass/Fail)	Pass
Tested By / Actor	Admin

Test case: Manage Users (Activate / Deactivate)

Field	Details
Test Case ID	TC-20
Use Case Name	UC-20 – Manage Users
Description	Verify admin can activate or deactivate a user

Input	User ID, active/inactive status
Expected Output	User status updated successfully
Result (Pass/Fail)	Pass
Tested By / Actor	Admin

Test case: Remove Report Media

Field	Details
Test Case ID	TC-21
Use Case Name	UC-21 – Remove Report Media
Description	Verify admin can remove inappropriate media
Input	Media ID
Expected Output	Media deleted from storage and database
Result (Pass/Fail)	Pass
Tested By / Actor	Admin

Test case: Send Push Notification (System)

Field	Details
Test Case ID	TC-22
Use Case Name	UC-22 – Send Push Notification
Description	Verify system sends notifications on events
Input	Disaster report or SOS trigger
Expected Output	Push notification delivered to users
Result (Pass/Fail)	Pass
Tested By / Actor	System (Firebase Cloud Messaging)

Test case: Calculate Report Verification (System)

Field	Details
Test Case ID	TC-23
Use Case Name	UC-23 – Calculate Report Verification
Description	Verify system calculates verification based on votes
Input	Multiple user votes
Expected Output	Report verification status updated correctly
Result (Pass/Fail)	Pass
Tested By / Actor	System (Cloud Functions)

Test case: Update User Reputation (System)

Field	Details
Test Case ID	TC-24
Use Case Name	UC-24 – Update User Reputation
Description	Verify system updates user reputation scores
Input	Verified/false reports, votes
Expected Output	User reputation updated accurately
Result (Pass/Fail)	Pass
Tested By / Actor	System (Cloud Functions)

Test case: Store Media in Firebase Storage

Field	Details
Test Case ID	TC-25
Use Case Name	UC-25 – Store Media
Description	Verify system stores uploaded media securely

Input	Image or video file
Expected Output	Media stored and URL saved in database
Result (Pass/Fail)	Pass
Tested By / Actor	System

Test case: Track User Location (GPS)

Field	Details
Test Case ID	TC-26
Use Case Name	UC-26 – Track User Location
Description	Verify system captures user GPS location
Input	Device GPS coordinates
Expected Output	Location captured accurately
Result (Pass/Fail)	Pass
Tested By / Actor	System